The Diamond Model

Note: This content will help you once you have gone through the book. If you did not read this topic from the book before, you may face some trouble. In that case, check that particular topic from the book or simply ask someone to explain the issue who have the habit of reading books. Also do not use any unfair means in the exam using this content (or any content).

[Most of the credit of this content goes to Tushar Siddik and Vaskor Dey Dhrubo]

The Diamond Model

- Developed by Diamond (1965).
- This is the overlapping-generations model.

2.8 Assumptions

- The central difference between the Diamond model and the RCK model
  - There is turnover in the population:
  - New individuals are continually being born, and
  - Old individuals are continually dying.
- Time is discrete (t = 0,1,2,...) rather than continuous.
- Each individual **lives for only two periods.**

- Individuals are born in period t: $\boldsymbol{L_t}$
- Population grows at rate $\boldsymbol{n}$
- Thus, $\boldsymbol{L_t = (1+n)L_{t-1}}$

- At time *t*:
  - Individuals in the **first period** of their lives: $\boldsymbol{L_t}$
  - Individuals in their **second periods**: $L_{t-1} = \frac{\boldsymbol{L_t}}{\boldsymbol{1+n}}$

- Each individual supplies 1 unit of labor **when he or she is young** and divides the resulting labor **income** between **first-period consumption** and **saving**.
- In the **second period**, the individual simply **consumes** the **saving** and any **interest** he or she earns.

- **Consumption** in period *t* of **young** individuals: $\boldsymbol{C_{1t}}$
- **Consumption** in period *t* of **old** individuals: $\boldsymbol{C_{2t}}$

- We again assume constant-relative-risk-aversion

$$\boldsymbol{U_t = \frac{C_{1t}^{1-\theta}}{1-\theta} + \frac{1}{1+\rho}\frac{C_{2t+1}^{1-\theta}}{1-\theta}} \qquad \theta > 0, \rho > -1 \qquad (2.43)$$

  - If $\boldsymbol{\rho > 0}$, individuals place **greater weight on first-period** than second-period consumption.
  - If $\boldsymbol{\rho < 0}$, individuals place **greater weight on second-period** consumption.
  - The assumption $\boldsymbol{\rho > -1}$ ensures that the **weight on second period consumption is positive**.

Assumptions about farms are same as RCK model.

- Production function Y_t = F (K_t, A_tL_t)
- F (•) again has **constant returns** to scale and satisfies the **Inada conditions**
- **A** again grows at exogenous rate ***g*** $[A_t = (1+g)A_{t-1}]$
- Markets are **competitive**
- There is **no depreciation**
- The real interest rate $r_t = f'(k_t)$
- Wage per unit of effective labor $w_t = f(k_t) - k_tf'(k_t)$.

(All the above assumptions are same as RCK model)

- Initial capital stock, $\boldsymbol{K_0}$
  - Strictly positive
  - Owned equally by all **old individuals**

Thus, in period *0* the capital owned by the old and the labor supplied by the young are combined to produce output.

- The capital stock in period *t* +*1*:

$$K_{t+1} = L_t(w_tA_t - C_{1t}) \tag{2.44}$$

  - L_t = number of young individuals in period *t*
  - $(w_tA_t - C_{1t})$ = individuals' saving

2.9 Household Behavior

The second-period consumption of an individual born at *t* is

$$C_{2t+1} = (1+ r_{t+1})(w_tA_t - C_{1t}) \tag{2.45}$$

Dividing both sides of this expression by 1+ r_{t+1} and bringing C_{1t} over to the left -hand side yields the individual's **budget constraint**:

$$\boldsymbol{C_{1t} + \frac{1}{1+r_{t+1}}C_{2t+1} = A_tw_t} \tag{2.46}$$

This condition states that **the present value of lifetime consumption** equals **initial wealth** (which is zero) plus the **present value of lifetime labor income** (which is A_tw_t)

  - The individual maximizes utility, (2.43), subject to the budget constraint, (2.46).

- We will consider two ways of solving this maximization problem.

  i) Proceeding along the lines of the **intuitive derivation of the Euler equation** for the RCK model
  ii) Setting up **Lagrangian**

i) ***Proceeding along Euler equation derivation***

- We imagine the individual decreasing C_{1t} by a small amount, ΔC
- Then using the additional saving and capital income to raise C_{2t+1} by $(1+r_{t+1})\Delta C$

If the individual is optimizing, the **utility cost** and **benefit of the change** must be equal.

- The marginal contributions of C_{1t} to lifetime utility is $C_{1t}^{-\theta}$
- The marginal contributions of C_{2t+1} to lifetime utility is $\frac{1}{1+\rho}\, C_{2t+1}^{-\theta}$

Thus as we let ΔC approach 0,

- The **utility cost** of the change approaches $C_{1t}^{-\theta}\ \Delta C$
- The **utility benefit** approaches $\frac{1}{1+\rho}\ C_{2t+1}^{-\theta}(1+r_{t+1})\Delta C$

As just described, these are equal when the individual is optimizing. Thus, optimization requires

$$C_{1t}^{-\theta}\Delta C = \frac{1}{1+\rho}\, C_{2t+1}^{-\theta}(\ 1+r_{t+1})\ \Delta C \tag{2.47}$$

Canceling the ΔC's and multiplying both sides by C_{2t+1}^{θ} gives us

$$\frac{C_{2t+1}^{\theta}}{C_{1t}^{\theta}} = \frac{1+r_{t+1}}{1+\rho} \tag{2.48}$$

Or

$$\frac{C_{2t+1}}{C_{1t}} = \left(\frac{1+r_{t+1}}{1+\rho}\right)^{\frac{1}{\theta}} \tag{2.49}$$

- (This can be treated as the **Euler Equation** of the Diamond Model)
- This condition and the budget constraint describe the individual's behavior.
- It implies that whether an individual's consumption is **increasing** or **decreasing** over time depends on whether the **real rate of return** is **greater** or **less** than the **discount rate**.

ii) ***Setting up Lagrangian***

$$\mathcal{L} = \frac{C_{1t}^{1-\theta}}{1-\theta} + \frac{1}{1+\rho}\frac{C_{2t+1}^{1-\theta}}{1-\theta} + \lambda\left(A_t w_t - \left(C_{1t} + \frac{C_{2t+1}}{1+r_{t+1}}\right)\right) \tag{2.50}$$

- **FOC** for C_{1t} is

$$C_{1t}^{-\theta} = \lambda \tag{2.51}$$

- **FOC** for C_{2t+1} is

$$\frac{1}{1+\rho}C_{2t+1}^{-\theta} = \frac{\lambda}{1+r_{t+1}} \tag{2.52}$$

Substituting the first equation into the second yields

$$\frac{1}{1+\rho} C_{2t+1}^{-\theta} = \frac{C_{1t}^{-\theta}}{1+r_{t+1}} \tag{2.53}$$

$$\Rightarrow \frac{1+r_{t+1}}{1+\rho} = \frac{C_{2t+1}^{\theta}}{C_{1t}^{\theta}}$$

$$\Rightarrow \frac{C_{2t+1}}{C_{1t}} = \left(\frac{1+r_{t+1}}{1+\rho}\right)^{\frac{1}{\theta}}$$

This is same as (2.49).

As before, this condition and the budget constraint characterize utility-maximizing behavior.

(The following part is bit complex and tough to relate)

We can use the Euler equation, (2.49), and the budget constraint, (2.46), to express C_{1t} in terms of labor income and the real interest rate.

Multiplying both sides of (2.49) by C_{1t} gives

$$C_{2t+1} = \left(\frac{1+r_{t+1}}{1+\rho}\right)^{\frac{1}{\theta}} C_{1t}$$

We substitute this value into (2.46)

$$C_{1t} + \frac{\left(\frac{1+r_{t+1}}{1+\rho}\right)^{\frac{1}{\theta}} C_{1t}}{1+r_{t+1}} = A_t w_t$$

$$\Rightarrow C_{1t} + \frac{\frac{(1+r_{t+1})^{\frac{1}{\theta}}}{(1+\rho)^{\frac{1}{\theta}}}}{1+r_{t+1}} C_{1t} = A_t w_t$$

$$\Rightarrow C_{1t} + \frac{(1+r_{t+1})^{\frac{1}{\theta}-1}}{(1+\rho)^{\frac{1}{\theta}}} C_{1t} = A_t w_t$$

$$\Rightarrow C_{1t} + \frac{(1+r_{t+1})^{\frac{1-\theta}{\theta}}}{(1+\rho)^{\frac{1}{\theta}}} C_{1t} = A_t w_t \tag{2.54}$$

$$\Rightarrow C_{1t}\left(1 + \frac{(1+r_{t+1})^{\frac{1-\theta}{\theta}}}{(1+\rho)^{\frac{1}{\theta}}}\right) = A_t w_t$$

$$\Rightarrow C_{1t}\left(\frac{(1+\rho)^{\frac{1}{\theta}} + (1+r_{t+1})^{\frac{1-\theta}{\theta}}}{(1+\rho)^{\frac{1}{\theta}}}\right) = A_t w_t$$

$$\Rightarrow C_{1t} = \frac{(1+\rho)^{\frac{1}{\theta}}}{(1+\rho)^{\frac{1}{\theta}} + (1+r_{t+1})^{\frac{1-\theta}{\theta}}} A_t w_t \tag{2.55}$$

Equation (2.55) shows that the **interest rate** determines **the fraction of income the individual consumes** in the first period.

If we let *s(r)* denote the **fraction of income saved**, (2.55) implies

$$s(r) = \frac{(1+r)^{\frac{1-\theta}{\theta}}}{(1+\rho)^{\frac{1}{\theta}}+(1+r)^{\frac{1-\theta}{\theta}}} \tag{2.56}$$

We can therefore rewrite (2.55) as

$$C_{1t} = [1 - s(r)]A_t w_t \tag{2.57}$$

Equation (2.55) implies that

- young individuals' saving is
  - Increasing in r if **θ < 1**
  - Decreasing if **θ > 1**

- A rise in *r* has both an income and a substitution effect.

  - **<u>The substitution effect</u>**: If **second-period consumption** become **more favorable**, first period consumption deceases and **saving increases**.

  - **<u>The income effect:</u>** If a given amount of saving yields more second-period consumption then first period consumption increases and **saving decreases**.

  - When ***θ* is low**, the **substitution effect** dominates
  - When ***θ* is high**, the **income effect** dominates.
  - And in the special case of ***θ* =1** (**logarithmic utility**), the **two effects balance**

(In my judgement, the chance is very little that there will be any question from the above portion. I think all this discussion is done in order to introduce us to the idea of **logarithmic utility.** But you have all the right to disagree with me.)

2.10 The Dynamics of the Economy

The Equation of Motion of k

- The capital stock in period *t +1* is the amount saved by young individuals in period *t*.
- Since the young save fraction $s(r_{t+1})$ of their labor income, this implies

$$K_{t+1} = s(r_{t+1})L_t A_t w_t \tag{2.58}$$

- Dividing both sides of (2.58) by $L_{t+1}A_{t+1}$ gives us:

$$k_{t+1} = \frac{1}{(1+n)(1+g)} s(r_{t+1}) w_t \tag{2.59}$$

Substituting for $r_{t+1} = f'(k_{t+1})$ and $w_t = f(k_t) - k_t f'(k_t)$:

$$\boldsymbol{k_{t+1} = \frac{1}{(1+n)(1+g)} s\big(f'(k_{t+1})\big)[f(k_t) - k_t f'(k_t)]} \tag{2.60}$$

The Evolution of k

- Equation (2.60) implicitly defines $\boldsymbol{k_{t+1}}$ **as a function of** $\boldsymbol{k_t}$
- A value of k_t such that $\boldsymbol{k_{t+1}} = \boldsymbol{k_t}$ satisfies (2.60) **is a BGP value of *k***

  - We therefore want to know whether there is a BGP value (or values) of *k*, and
  - Whether *k* converges to such a value if it does not begin at one.

- To answer these questions, we begin by considering the case of **logarithmic utility** and **Cobb Douglas production.**

Logarithmic Utility and Cobb–Douglas Production

- **When $\boldsymbol{\theta = 1}$**, the fraction of labor income **saved** is $\boldsymbol{\frac{1}{2+\rho}}$
- When production is Cobb–Douglas
  - *f(k)* $= k^{\alpha}$ and
  - $f'(k) = \alpha k^{\alpha-1}$

Equation (2.60) therefore becomes

$$\boldsymbol{k_{t+1} = \frac{1}{(1+n)(1+g)} \frac{1}{2+\rho} (1-\alpha) k_t^{\alpha}} \tag{2.61}$$

Explanation:

$$k_{t+1} = \frac{1}{(1+n)(1+g)} \frac{1}{2+\rho} [\boldsymbol{k_t^{\alpha} - k_t\, \alpha k_t^{\alpha-1}}]$$

$$= \frac{1}{(1+n)(1+g)} \frac{1}{2+\rho} \left[k_t^{\alpha} - k_t \frac{\alpha k_t^{\alpha}}{k_t}\right]$$

$$= \frac{1}{(1+n)(1+g)} \frac{1}{2+\rho} [k_t^{\alpha} - \alpha k_t^{\alpha}]$$

$$k_{t+1} = \frac{1}{(1+n)(1+g)} \frac{1}{2+\rho} (1-\alpha) k_t^{\alpha}$$

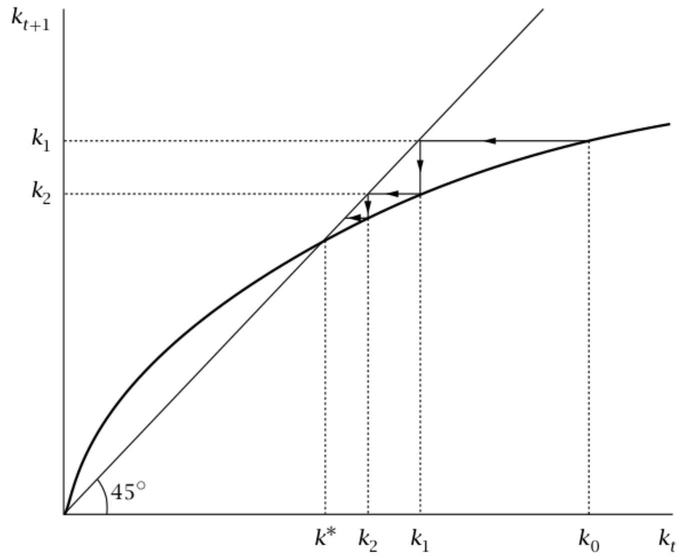

FIGURE 2.11 The dynamics of k

Figure 2.11 shows k_{t+1} as a function of k_t.

- A point where the $\boldsymbol{k_{t+1}}$ **function intersects the 45-degree line** is a point where $\boldsymbol{k_{t+1} = k_t}$.

- In the case we are considering k_{t+1} equals k_t at $k_t = 0$;
  - It rises above k_t when $\boldsymbol{k_t}$ **is small**; and
  - It then crosses the **45-degree line** and remains below.

- There is thus **a unique BGP level** of k (aside from k = 0), which is denoted $\boldsymbol{k}^*$.
- $\boldsymbol{k}^*$ is **globally stable**: wherever k starts **it converges to $\boldsymbol{k}^*$**

- The **properties** of the economy once it has converged to its **BGP** are the same as those of the Solow and Ramsey economies on their BGPs:
  - The saving rate is constant,
  - Output per worker is growing at rate g,
  - The capital-output ratio is constant, and so on.

Effect of fall in the discount rate

- The economy is initially on its BGP and there is a **fall in the discount rate**, $\boldsymbol{\rho}$
  - The **fall in $\boldsymbol{\rho}$** causes the young to **save a greater fraction** of their labor income.
  - Thus the $\boldsymbol{k_{t+1}}$ function **shifts up.**

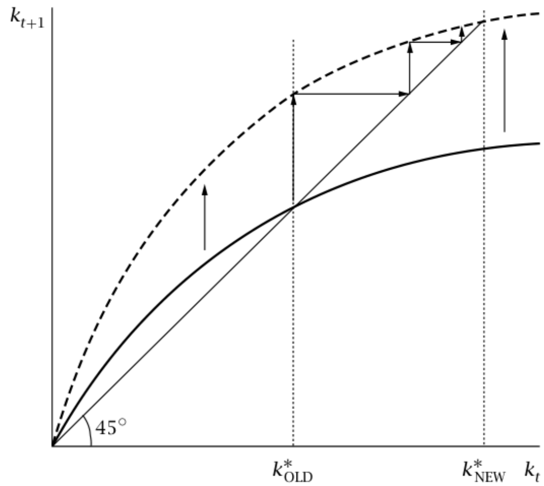

FIGURE 2.12 The effects of a fall in the discount rate

This is depicted in Figure 2.12.

- The **upward shift** of the $\boldsymbol{k_{t+1}}$ function **increases $\boldsymbol{k^*}$**, the value of *k* on the BGP.

- Thus, the effects of a fall in the discount rate in the Diamond model in the case we are considering are
  - Similar to its effects in the RCK model, and
  - To the effects of a rise in the saving rate in the Solow model.

- The change **shifts the paths over time** of **output and capital per worker permanently up**, but it leads only to temporary increases in the **growth rates** of these variables.

The Speed of Convergence

- Equation (2.61) gives $\boldsymbol{k_{t+1}}$ as a function of $\boldsymbol{k_t}$.
- The economy is on its BGP when these two are equal. That is, $\boldsymbol{k^*}$ is defined by

$$k^* = \frac{1}{(1+n)(1+g)}\frac{1}{2+\rho}(1-\alpha){k^*}^{\alpha} \tag{2.62}$$

Solving this expression for $\boldsymbol{k^*}$ yields

$$\boldsymbol{k^* = \left(\frac{1-\alpha}{(1+n)(1+g)(2+\rho)}\right)^{\frac{1}{1-\alpha}}} \tag{2.63}$$

Since $y^* = {k^*}^{\alpha}$:

$$y^* = \left(\frac{1-\alpha}{(1+n)(1+g)(2+\rho)}\right)^{\frac{\alpha}{1-\alpha}} \tag{2.64}$$

This expression shows how the model's parameters affect output per unit of effective labor on the BGP.

We can also find how quickly the economy converges to the BGP

To do this,

- We again **linearize** around BGP.
- That is, we replace the equation of motion for k, (2.61), with a first order approximation around $\mathbf{k} = \boldsymbol{k}^*$**.**

We know that when $\boldsymbol{k_t} = \boldsymbol{k}^*$, $\boldsymbol{k_{t+1}}$ also equals $\boldsymbol{k}^*$. Thus,

$$k_{t+1} \simeq k^* + \left(\frac{dk_{t+1}}{dk_t}\bigg|_{k_t = k^*}\right)(k_t - k^*) \tag{2.65}$$

Let, $\boldsymbol{\lambda} = \frac{\boldsymbol{dk_{t+1}}}{\boldsymbol{dk_t}}$ evaluated at $k_t = k^*$. With this definition we can rewrite (2.65) as

$$k_{t+1} - k^* \simeq \lambda(k_t - k^*)$$

This implies

$$k_t - k^* = \boldsymbol{\lambda^t}(\boldsymbol{k_0} - k^*) \tag{2.66}$$

where k_0 = the initial value of k.

Explanation:

$$k_{t+1} - k^* \simeq \lambda(k_t - k^*)$$

$$k_t - k^* = \frac{1}{\lambda}(k_{t+1} - k^*)$$

$$= \boldsymbol{\lambda^{-1}}(k_{\boldsymbol{t+1}} - k^*)$$

$$= \boldsymbol{\lambda}^{-(-t)}(\boldsymbol{k_{t-t}} - k^*)$$

$$= \boldsymbol{\lambda^t}(\boldsymbol{k_0} - k^*)$$

- The convergence to the BGP is **determined by** λ.
  - If λ is between **0 and 1**, the system **converges smoothly**.

To find λ, we return to (2.61): $k_{t+1} = \frac{1}{(1+n)(1+g)}\frac{1}{2+\rho}(1-\alpha)k_t^{\alpha}$

Thus,

$$\lambda = \left(\frac{dk_{t+1}}{dk_t}\bigg|_{k_t = k^*}\right) = \alpha\frac{1-\alpha}{(1+n)(1+g)(2+\rho)}{k^*}^{\alpha-1}$$

$$= \alpha\frac{1-\alpha}{(1+n)(1+g)(2+\rho)}\left(\frac{1-\alpha}{(1+n)(1+g)(2+\rho)}\right)^{\frac{\alpha-1}{1-\alpha}} \quad \text{[using the value of } k^* \text{ from equation (2.63)]}$$

$$= \alpha \tag{2.67}$$

Explanation:

$$\lambda = \left(\frac{dk_{t+1}}{dk_t}\bigg|_{k_t = k^*}\right)$$

$$= \frac{d}{dk_t}\left[\frac{1-\alpha}{(1+n)(1+g)(2+\rho)}{k^*}^{\alpha}\right]$$

$$= \alpha\frac{1-\alpha}{(1+n)(1+g)(2+\rho)}{k^*}^{\alpha-1}$$

$$= \alpha\frac{1-\alpha}{(1+n)(1+g)(2+\rho)}\left[\left(\frac{1-\alpha}{(1+n)(1+g)(2+\rho)}\right)^{\frac{1}{1-\alpha}}\right]^{\alpha-1}$$ [using k^* from equation (2.63)]

$$= \alpha\frac{1-\alpha}{(1+n)(1+g)(2+\rho)}\left(\frac{1-\alpha}{(1+n)(1+g)(2+\rho)}\right)^{\frac{\alpha-1}{1-\alpha}}$$

$$= \alpha\frac{1-\alpha}{(1+n)(1+g)(2+\rho)}\left(\frac{1-\alpha}{(1+n)(1+g)(2+\rho)}\right)^{-1}$$

$$= \alpha\frac{(1-\alpha)}{(1+n)(1+g)(2+\rho)}\frac{(1+n)(1+g)(2+\rho)}{(1-\alpha)}$$

$$= \alpha$$

- That is, λ is simply α, **capital's share**.
- Since **α is between 0 and 1**, this analysis implies that ***k* converges smoothly to *k^**.**
  - For example: If α is one-third, k moves two-thirds of the way toward k^* each period.

- The rate of convergence in the Diamond model differs from that in the Solow model.

  - Because **total saving** as a fraction of output is a decreasing function of k.

- Thus, **total saving** as a fraction of output is
  - **Above** its BGP value when **k** < k^*, and
  - **Below** when **k** > k^*.

As a result, **convergence is more rapid** than in the Solow model.

The General Case

Let us now relax the assumptions of logarithmic utility and Cobb–Douglas production..

We can rewrite the equation of motion, (2.60), as

$$k_{t+1} = \frac{1}{(1+n)(1+g)}s\big(f'(k_{t+1})\big)\frac{f(k_t)-k_t f'(k_t)}{f(k_t)}f(k_t) \tag{2.68}$$

Figure 2.13 shows some possible forms for the relation between k_{t+1} and k_t other than the well-behaved case shown in Figure 2.11.

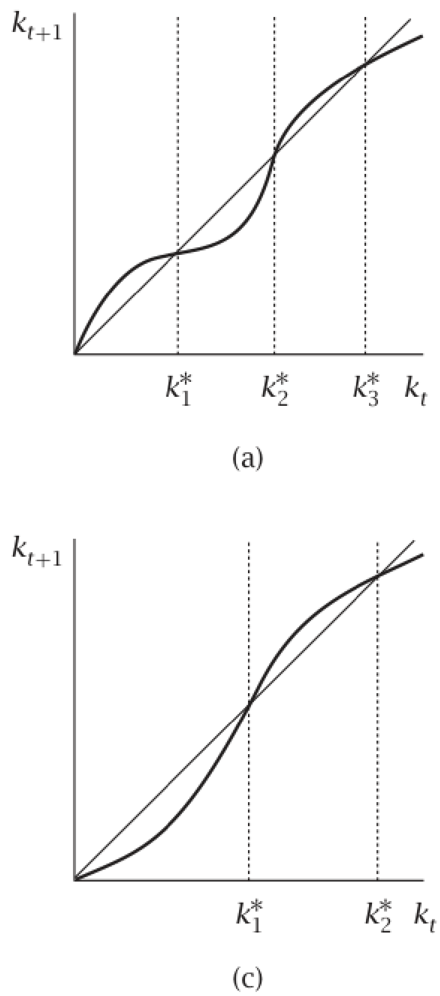

FIGURE 2.13 Various possibilities for the relationship between k_t and k_{t+1}

Panel (a) shows a case with multiple values of k^*

- In the case shown, k_1^* and k_3^* are **stable:**
  - If k starts **slightly away** from one of these points, **it converges** to that level.

- k_2^* is **unstable** (as is k = 0).
  - If k starts slightly below k_2^*, then k_{t+1} is less than k_t each period, and so $\boldsymbol{k}$ **converges to $\boldsymbol{k_1^*}$.**
  - If k begins slightly above k_2^*, it **converges to $\boldsymbol{k_3^*}$.**

Panel (b) shows a case in which k_{t+1} is always less than k_t

- k therefore converges to zero regardless of its initial value.

Panel (c) shows a case in which

- $\boldsymbol{k}$ **converges to zero** if its **initial value is sufficiently low**, but
- To a **strictly positive level** if its **initial value is sufficiently high**

- Specifically, if $\boldsymbol{k_0 < k_1^*}$, then $\boldsymbol{k}$ **approaches zero**;
- If $\boldsymbol{k_0 > k_1^*}$, then $\boldsymbol{k}$ **converges to** $\boldsymbol{k_2^*}$**.**

Finally, **Panel (d) shows a case** in which

- k_{t+1} is not **uniquely determined** by k_t:
  - When k_t is **between $\boldsymbol{k_a}$ and $\boldsymbol{k_b}$,** there are **three possible values of $\boldsymbol{k_{t+1}}$**.
- Thus the path of the economy is **indeterminate:**
  - Equation (2.60) (or [2.68]) **does not fully determine how $\boldsymbol{k}$ evolves over time** given its initial value.
  - This raises the possibility that **self-fulfilling prophecies** and **sunspots** can affect the **behavior of the economy**
  - and that the economy can exhibit **fluctuations** even though there **are no exogenous disturbances**.

Once again, **growth in the effectiveness of labor** is the only potential source of long-run growth in output per worker.

Because of the possibility of multiple k^*'s, the model does imply that otherwise identical economies can converge to **different BGPs** simply because of differences in their initial conditions.

2.11 The Possibility of Dynamic Inefficiency

- The one major difference between the BGP of the Diamond and RCK models involves **welfare**.
- We saw that the equilibrium of **the RCK model maximizes the welfare** of the representative household.
- That is, equilibrium of RCK model is Pareto-efficient.
- But the equilibrium of the **Diamond model need not to satisfy Pareto-efficiency**.
- In particular, the **capital stock** on the BGP of the Diamond model **may exceed** the **golden-rule level**
  - So that a **permanent increase in consumption** is possible.

To see this possibility as simply as possible, we assume that

- ➢ Utility is logarithmic
- ➢ Production is Cobb–Douglas, and
- ➢ g is zero.

With $g = 0$, equation (2.63) for the value of k on the BGP simplifies to

$$k^* = \left(\frac{1-\alpha}{(1+n)(2+\rho)}\right)^{\frac{1}{1-\alpha}} \qquad (2.69)$$

The marginal product of capital on the BGP, $f'(k^*) = \boldsymbol{\alpha(k^*)^{\alpha-1}}$ is

$$f'(k^*) = \frac{\alpha}{(1-\alpha)}(1+n)(2+\rho) \qquad (2,70)$$

Explanation:

$$f'(k^*) = \alpha(k^*)^{\alpha-1} = \alpha\left(\left(\frac{1-\alpha}{(1+n)(2+\rho)}\right)^{\frac{1}{1-\alpha}}\right)^{\alpha-1}$$

$$= \alpha\left(\frac{1-\alpha}{(1+n)(2+\rho)}\right)^{\frac{\alpha-1}{1-\alpha}}$$

$$= \alpha\left(\frac{1-\alpha}{(1+n)(2+\rho)}\right)^{-1}$$

α

- On a BGP with g= 0, **total consumption per unit of effective labor =** ***f*(*k*) − *nk***
- The golden-rule capital stock therefore satisfies $\boldsymbol{f'(k_{GR}) = n}$.
- $f'(k^*)$ can be either more or less than $f'(k_{GR})$.
- In particular, **for α sufficiently small**, $f(k^*)$ is less than $f(k_{GR})$
    - ➢ *That is,* **the capital stock** on the BGP **exceeds** the **golden-rule level**.

Why $k^* > k_{GR}$ is inefficient:

- We introduce a social planner into a Diamond economy that is on its BGP with $\boldsymbol{k^* > k_{GR}}$.
- The amount of **output per worker** available each period for **consumption** = $f(k^*) - nk^*$.
- This is shown by the crosses in Figure 2.14.

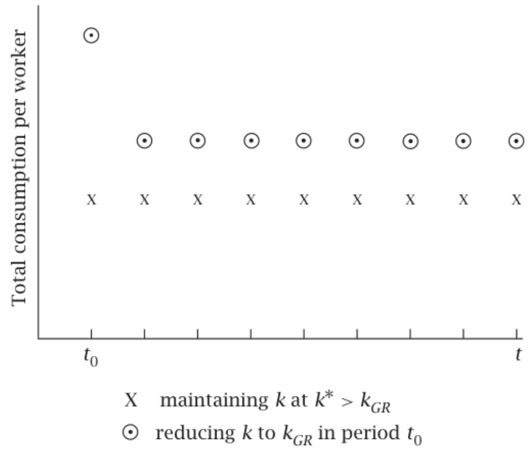

FIGURE 2.14 How reducing k to the golden-rule level affects the path of consumption per worker

Suppose instead, however, that in some period, period t_0,

- The planner allocates **more resources to consumption** and **fewer to saving** than usual
  - So that capital per worker the **next period** is $\boldsymbol{k_{GR}}$
  - Thereafter he or she **maintains $\boldsymbol{k}$ at $\boldsymbol{k_{GR}}$.**

- Under this plan, the **resources per worker** available for consumption
  - In period t_0 are $\boldsymbol{f(k^*)+(k^*-k_{GR})-nk_{GR}}$.
  - In each subsequent period
    the **output per worker** available for consumption is $\boldsymbol{f(k_{GR})-nk_{GR}}$.

Since k_{GR} maximizes $f(k)-nk$

- $\boldsymbol{f(k_{GR})-nk_{GR}}$ exceeds $\boldsymbol{f(k^*)-nk^*}$.

And since k^* is greater than k_{GR},

- $\boldsymbol{f(k^*)+(k^*-k_{GR})-nk_{GR}}$ rise **even larger** than $\boldsymbol{f(k_{GR})-nk_{GR}}$.

- The path of total consumption **under this policy** is shown by the circles in Figure 2.14.

As the figure shows,

- This policy makes **more resources available for consumption** in every period than the policy of maintaining k at k^*.
- The planner can therefore allocate consumption between the young and the old each period to make **every generation better off**.

Thus, the equilibrium of the Diamond model can be Pareto-inefficient

The reason is that the standard result assumes **not only** competition and an absence of externalities, **but also** a **finite number of agents**.

- Specifically, the possibility of inefficiency in the Diamond model stems from the fact that
  - The **infinity of generations** gives the planner a **means of providing** for the **consumption** of **the old** that is not available to the market.

Because this type of inefficiency

- Differs from **conventional sources** of inefficiency,
- Stems from the **intertemporal structure** of the economy

it is known as **dynamic inefficiency.**

2.12 Government in the Diamond Model

For simplicity, we focus on the case of **logarithmic utility** and **Cobb–Douglas production**.

- Let $\boldsymbol{G_t}$ denote the **government's purchases** of goods per unit of effective labor in period *t*.
- Assume that it **finances** those purchases by **lump-sum taxes on the young**.
  - Workers' after-tax income in period *t* is $(\mathbf{1}-\boldsymbol{\alpha})\boldsymbol{k_t^{\alpha}}-\boldsymbol{G_t}$
  - The equation of motion for *k*, equation (2.61), therefore becomes

$$\boldsymbol{k_{t+1}}=\frac{\mathbf{1}}{(\mathbf{1}+\boldsymbol{n})(\mathbf{1}+\boldsymbol{g})(\mathbf{2}+\boldsymbol{\rho})}[(\mathbf{1}-\boldsymbol{\alpha})\boldsymbol{k_t^{\alpha}}-\boldsymbol{G_t}] \tag{2.71}$$

A **higher $\boldsymbol{G_t}$** therefore reduces k_{t+1} for a given k_t.

Effects of Permanent Increase in Government Purchases:

- Suppose that the economy is on a BGP with G constant; and that
- G increases **permanently.**
  - From (2.71), this **shifts the $\boldsymbol{k_{t+1}}$ function down**;
  - This is shown in Figure 2.15.
  - The downward shift of the k_{t+1} function **reduces $\boldsymbol{k^*}$**.

Thus—in contrast to what occurs in the infinite-horizon model (RCK Model)

- **Higher government purchases** lead to a **lower capital stock** and a **higher real interest rate**.

Effects of Temporary Increase in Government Purchases:

- The economy initially on its BGP
- There is **a temporary increase** in government purchases from $\boldsymbol{G_L}$ to $\boldsymbol{G_H}$

The dynamics of k are thus

- Described by (2.71) with **G =** $\boldsymbol{G_H}$ during the period that government purchases are high and
- By (2.71) with **G =** $\boldsymbol{G_L}$ before and after.

That is, the fact that individuals know that government purchases will return to G_L **does not affect the behavior of the economy** during the time that purchases are **high**.

- The saving of the young— and hence next period's capital stock—is determined by their **after-tax labor income**,
  - Which is determined by the current capital stock and by the government's current purchases.

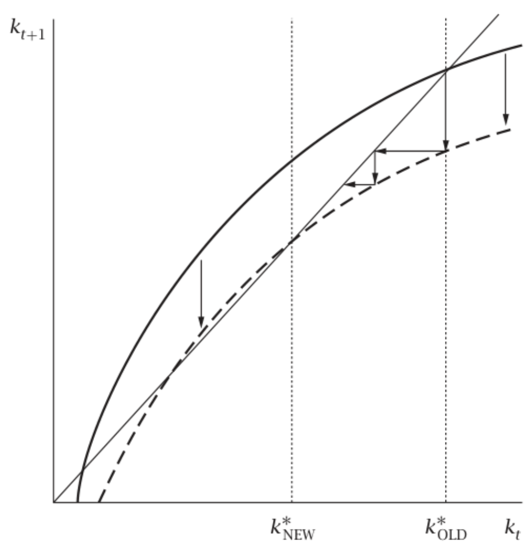

FIGURE 2.15 The effects of a permanent increase in government purchases

- Thus, during the time that government purchases are high,
  - k gradually falls and r gradually increases.
- Once G returns to G_L,
  - k rises gradually back to its initial level.

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